

Daniel Sánchez-García

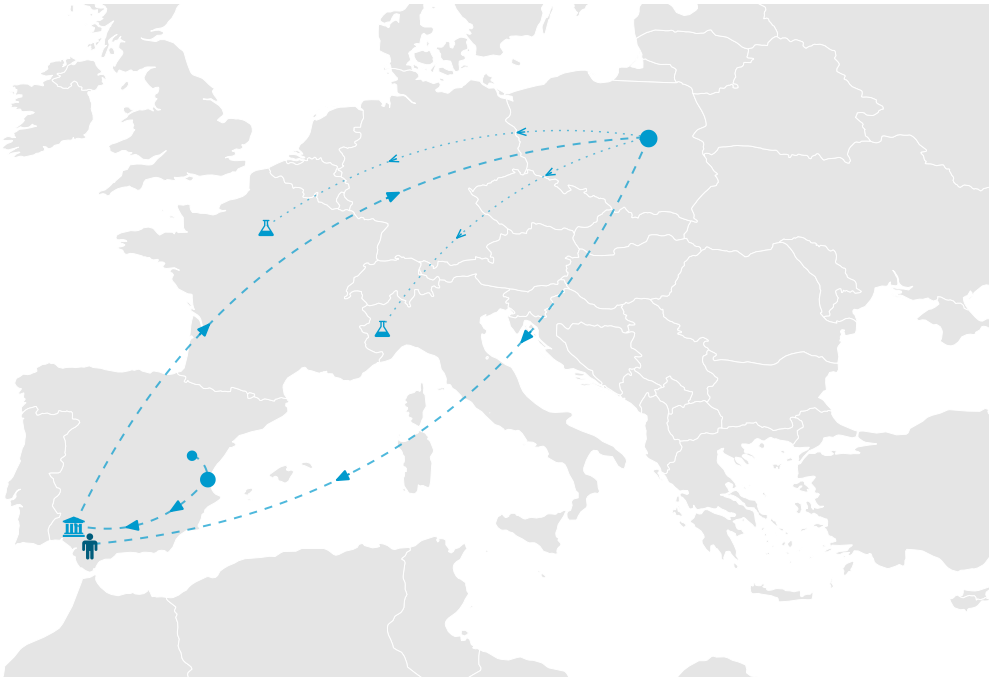
PhD in Biology

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My journey



Professional experience

Postdoctoral Researcher

Doñana Biological Station (EBD-CSIC), Department of Conservation Biology and Global Change, Seville (Spain) — Jan 2025 - present

Specialist in ant species identification

Doñana Biological Station (EBD-CSIC), Department of Conservation Biology and Global Change, Seville (Spain) — May 2024 - Jun 2024

- Identification of ant species from pitfall trap samples from Spain, France, Belgium and the Netherlands, contributing to ecological research and biodiversity assessments through taxonomic expertise.

Education

PhD in Biology

Museum and Institute of Zoology, Polish Academy of Sciences, Warsaw (Poland) — Oct 2019 - Sep 2024

- PhD Thesis: The project aimed to get a better understanding of how reintroductions of parasites work, we performed a multi-level comparison of source and reintroduced populations of *Phengaris teleius* by studying 1) population genetics, 2) morphology of adult butterflies and 3) behavioural, chemical and acoustical adaptations of butterfly caterpillar to the *Myrmica* host ants.

MSc in Biodiversity and Conservation Biology

Universidad Pablo de Olavide, Seville (Spain) — Oct 2017 - Jul 2018

- MSc Thesis (Thesis grade: 9.8/10 - Honors with Distinction): The work aimed to study the effect of temperature over ant assemblages from different habitats, applying different ecological perspectives. Taxonomic and functional diversity were studied, and a new index was developed to study the direct effect of temperature over the interspecific competition.

BSc in Biology

Universitat de València, Valencia (Spain) — Sep 2012 - Sep 2016

- BSc thesis (Thesis grade: 9.5/10): The work aimed to provide new knowledge about the ant communities of Sierra de Javalambre. The main goals were: 1) study how ant species are distributed along the altitudinal gradient, studying variables like ground temperature, nest density and stone density; 2) study the patching effect of *Juniperus sabina* over ant activity and nesting areas.

Research stays

Prof. Luca Pietro Casacci

University of Turin, Department of Life Sciences and Systems Biology, Turin (Italy) — Oct 2021

- I received training in recording insect vibroacoustic signals, sonogram analysis and performing behavioral assays, which enhanced my skills in experimental design and data interpretation.

Prof. Patrizia d'Ettorre

Sorbonne Paris North University, Laboratory of Experimental and Comparative Ethology, Paris (France) — Jan 2020 - Feb 2020

- I received training in GC-MS analysis of cuticular hydrocarbons and bioinformatics workflows, which deepened my understanding of chemical communication in social insects and enhanced my analytical skills.

Internships

Scientific assistant

Valencian Institute of Agricultural Research (IVIA), Department of Entomology, Valencia (Spain) — Feb 2016 - May 2016

- Internship with the researcher Alejandro Tena working on a biological control applied study about the effect of the presence of the ant *Lasius grandis* on the behaviour of the parasitoid wasp *Anagyrus pseudococci* on mealybug colonies of *Planococcus citri* (pest of citrus).

Environmental manager

Agricultural, Stockbreeding and Environmental Public Service, Government of Aragón, Teruel (Spain) — Jul 2015 - Sep 2015

- Internship with the environmental manager José Manuel González working on environmental impact reports; monitoring of fish river population dynamic for its management and conservation using electric fishing techniques; monitoring of deer population dynamic for its management and conservation in the National Hunting Reserve "Montes Universales"; management and translocation of native plant species.

Publications

Also in *my Google Scholar profile* and available to download from *my Website*.

Ambakina, H., Gómez, K., Monzenga, J.-C., **Sánchez-García, D.**, & Dekoninck, W. (2026). Uncovering ant diversity across forest successional stages in the yangambi biosphere reserve (DRC): Insights from winkler and pitfall trap sampling. *African Invertebrates*, *67*(1), 65–91. https://doi.org/10.3897/afrinvertebr.67.180461

Sánchez-García, D., Cerdá, X., & Angulo, E. (2026). Temperature and management reduce the abundance of the invasive argentine ant (Linepithema humile). *bioRxiv*. https://doi.org/10.64898/2026.10.08.756264

Sánchez-García, D., Wynhoff, I., d'Ettorre, P., Leroy, C., Kajzer-Bonk, J., Maák, I. E., Barbero, F., Casacci, L. P., & Witek, M. (2025). Ongoing coevolution between reintroduced Phengaris teleius butterflies and their Myrmica host ants. *bioRxiv*. https://doi.org/10.64898/2025.12.02.691951

Sánchez-García, D., Wynhoff, I., Kajzer-Bonk, J., Sztencel-Jabłonka, A., Nowicki, P., Casacci, L. P., & Witek, M. (2024). Temporal and spatial variation of morphological traits and genetic structure in Phengaris teleius myrmecophilous butterflies following habitat and climate changes three decades after reintroduction. *Global Ecology and Conservation*, *54*, e03104. https://doi.org/10.1016/j.gecco.2024.e03104

Trigos-Peral, G., Maák, I. E., Schmid, S., Chudzik, P., Czaczkes, T. J., Witek, M., Casacci, L. P., **Sánchez-García, D.**, Lőrincz, Á., Kochanowski, M., & Heinze, J. (2024). Urban abiotic stressors drive changes in the foraging activity and colony growth of the black garden ant Lasius niger. *Science of The Total Environment*, *915*, 170157. https://doi.org/10.1016/j.scitotenv.2024.170157

Sánchez-García, D., Cerdá, X., & Angulo, E. (2022). Temperature or competition: Which has more influence on mediterranean ant communities? *PLOS ONE*, *17*(4), e0267547. https://doi.org/10.1371/journal.pone.0267547

Angulo Aguado, E., Castro Cobo, S., **Sánchez-García, D.**, Sergio, F., Reyes-López, J. L., Álvarez Blanco, P., & Cerdá Sureda, X. (2019). ¿Se podrá controlar la expansión de la invasora hormiga argentina en doñana? In J. Junoy (Ed.), *Especies exóticas invasoras: Cátedra de parques nacionales* (pp. 249–262). Editorial Universidad de Alcalá.

García, F., Cuesta-Segura, A., Espadaler, X., García, J. C., & **Sánchez-García, D.** (2019). Lasius piliferus seifert, 1992: Descripción de la reina y actualización de su distribución ibérica (hymenoptera: formicidae). *Boletín de La Sociedad Entomológica Aragonesa (S.E.A.)*, *65*, 39–44.

García García, F., Espadaler Gelabert, X., Cuesta-Segura, A. D., & **Sánchez-García, D.** (2018). Primera cita ibérica para Temnothorax conatensis galkowski & lebas, 2016, y actualización de la distribución para Temnothorax grouvellei (bondroit, 1918) (hymenoptera: formicidae). *Iberomyrmex*, *10*, 22–27.

Espadaler, X., **Sánchez-García, D.**, & García-García, F. (2017). Temnothorax ibericus menozzi (1922), un endemismo ibérico orófilo (hymenoptera, formicidae). *Iberomyrmex*, *9*, 5–9.

Sánchez-García, D., & Espadaler, X. (2017). Una nueva especie parásita social para la península ibérica. Bothriomyrmex communista santschi, 1919 (hymenoptera: Formicidae) en españa. *Iberomyrmex*, *9*, 11–13.

Sánchez-García, D., & Espadaler, X. (2015). Cardiocondyla obscurior wheeler, 1929 (hymenoptera, formicidae) en españa. *Iberomyrmex*, *7*, 7–9.

Conference talks and posters

Sánchez-García, D., Stefanini, I., Lőrincz, Á., Trigos-Peral, G., Casacci, L. P., Ślipiński, P., Maák, I. E., Malewski, T., Oszako, T., & Witek, M. (2026). Ants shape the phyllosphere microbiome and drive tree health. *XX Iberian Congress of Myrmecology, Taxomara 2026*, *Móstoles*, Spain (Poster).

Sánchez-García, D., Taheri, A., Cerdá, X., & Angulo, E. (2025). ANT HUNTERS: A citizen science initiative for ant biodiversity research. *XIX Congress of the Spanish Society of Ethology and Evolutionary Ecology*, *Seville*, Spain (Poster).

Sánchez-García, D., Taheri, A., Cerdá, X., & Angulo, E. (2025). ANT HUNTERS: La ciudadanía a la caza de la biodiversidad de hormigas. *XIX International Congress of Myrmecology, Taxomara 2025*, *Girona*, Spain (Poster).

Sánchez-García, D., Wynhoff, I., d'Ettorre, P., Leroy, C., Kajzer-Bonk, J., Maák, I. E., Barbero, F., Casacci, L. P., & Witek, M. (2025). Mariposas mirmecófilas reescriben su lenguaje para adaptarse a nuevas hormigas huésped. *XIX International Congress of Myrmecology, Taxomara 2025*, *Girona*, Spain (Talk).

Sánchez-García, D., Wynhoff, I., d'Ettorre, P., Leroy, C., Kajzer-Bonk, J., Maák, I. E., Barbero, F., Casacci, L. P., & Witek, M. (2025). Reintroduction provides insight into multimodal mimicry evolution in an ant social parasite. *XIX Congress of the Spanish Society of Ethology and Evolutionary Ecology*, *Seville*, Spain (Talk).

Sánchez-García, D., Cerdá, X., & Angulo, E. (2024). Modificación del hábitat como herramienta de control de la invasión de la hormiga argentina. *XVIII International Congress of Myrmecology, Taxomara 2024*, *Málaga*, Spain (Talk).

Sánchez-García, D., Cerdá, X., & Angulo, E. (2024). Recovering native ant communities by removing suitable conditions for the invasive argentine ant. *IV International Young Researchers Conference on Invasive Species (IyrCIS)*, Online (Talk).

Sánchez-García, D., Casacci, L. P., Wynhoff, I., Kajzer-Bonk, J., Sztencel-Jabłonka, A., Nowicki, P., & Witek, M. (2023). Changes in morphology and genetic structure in two populations of Phengaris (= Maculinea) teleius 30 years after separation. *Butterfly Conservation Symposium 2023*, *Wyboston*, England (Talk).

Sánchez-García, D., Casacci, L. P., Wynhoff, I., Kajzer-Bonk, J., Nowicki, P., & Witek, M. (2022). Short-time evolution in the morphology of the myrmecophilous Maculinea teleius butterfly. *8th Polish Evolutionary Conference*, *Toruń*, Poland (Talk).

Sánchez-García, D. (2019). Fotografía de insectos: Material, técnicas y aplicaciones. *XIV International Congress of Myrmecology, Taxomara 2019*, *Chefchaouen*, Morocco (Talk).

Sánchez-García, D., Cerdá, X., & Angulo, E. (2018). ¿Temperatura o competencia, qué afecta más a las hormigas de doñana? *XIII Iberian Congress of Myrmecology, Taxomara 2018*, *León*, Spain (Talk).

Sánchez-García, D., Cuesta-Segura, A. D., Herraiz, J. A., Trigos-Peral, G., García García, F., Catarineu, C., Arcos González, J., & Fernández Martínez, J. A. (2017). Listado actualizado de las hormigas de la península ibérica e islas baleares (hymenoptera: formicidae). *XII Iberian Congress of Myrmecology, Taxomara 2017*, *Madrid*, Spain (Poster).

Sánchez-García, D. (2016). Aspectos ecológicos sobre la mirmecocenosis de solana de la sierra de javalambre (teruel). *XI Iberian Congress of Myrmecology, Taxomara 2016*, *Murcia*, Spain (Talk).

Funding

2022 - 2025	PRELUDIUM grant 2021/41/N/NZ8/04360 (NCN, Poland). Bacteria – ant – plant interaction: the effect of ants as bacteria dispersal vectors and indirect drivers of plant health.	*202,001 PLN (≈ 46,000 €)*

Awards

2025	Best oral presentation at the XIX International Congress of the Spanish Society of Ethology and Evolutionary Ecology (Seville, Spain)

Organization of Conferences and Scientific Events

2021	XVI International Conference of Myrmecology, Taxomara 2021	*Organizing Committee and Scientific Committee*
2020	XV International Conference of Myrmecology, Taxomara 2020	*Organizing Committee and Scientific Committee*

Reviews

Manuscripts reviewed for: Applied and Environmental Microbiology (1), Insect Conservation and Diversity (2), Arthropod-Plant Interactions (1), Polish Journal of Ecology (1)

Workshops

2025	Invited participant in InvaPact III: analysis of the ecological effects of biological invasions

Courses

2025	Systematic Reviews and Meta-analysis in R
2025	Optimizing the use of R for ecological analyses
2019	Advanced University Course in Ecological Network Analysis
2018	Animal Tracking Course

Memberships

since 2025	SORTEE - Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology
since 2012	Iberian Association of Myrmecology (member of the Governing Board)

Skills

Technical skills

Coding languages	R packages and tools	Other
R	R Markdown / Quarto – ggplot2 – vegan – geomorph	GC-MS – Morphometrics – Network analysis – Population genetics – Ant taxonomy – Oxford Nanopore sequencing – Macro photography

Languages

Skill	Spanish	English
Reading	Native	B2
Writing	Native	B2
Listening	Native	B2
Speaking	Native	B2

Common European Framework of Reference for Languages: A1/A2: Basic User. B1/B2: Independent User. C1/C2: Proficient User

References

- Postdoc: **Dra. Elena Angulo**, Doñana Biological Station - CSIC, Seville (Spain)
  angulo@ebd.csic.es
- PhD thesis: **Dra. Magdalena Witek**, Museum and Institute of Zoology, PAS, Warsaw (Poland)
  mwitek@miiz.waw.pl; mawitus@yahoo.co.uk
- Master's thesis: **Dr. Xim Cerdá**, Doñana Biological Station - CSIC, Seville (Spain)
  xim@ebd.csic.es
- Bachelor's thesis: **Prof. Dr. Ximo Baixeras**, University of Valencia, Valencia (Spain)
  joaquin.baixeras@uv.es